SE Lab 7: Fuzz Testing

Highway Toll Calculator: Lab Report

Name: Vismay Chandra Dev
SRN: PES2UG24CS597
Section: 5J
Assigned repo: github.com/SETAPESU26/L6_L7_36_highway_toll
My repo: github.com/VismayCD/L6_L7_36_highway_toll

1. Objective

To use the Hypothesis fuzzing library to find input validation bugs in highway_toll.py, then fix the code so that it handles unexpected or invalid input safely.

2. Running the Buggy Application

The original program works for valid input but crashes with a traceback on invalid input.

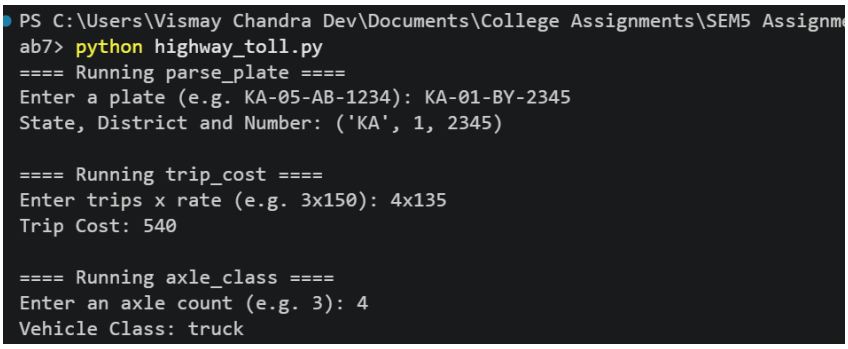


Figure 1: Buggy highway_toll.py with valid input.

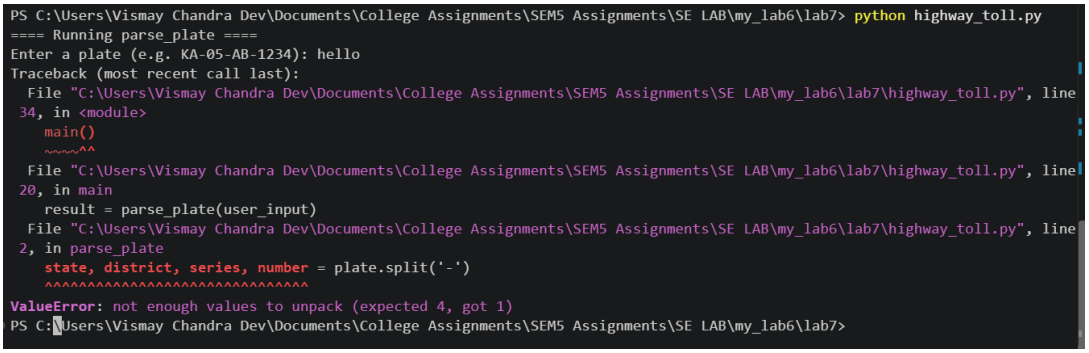


Figure 2: Buggy highway_toll.py crashing with a ValueError on the input "hello".

3. Fuzz Tests

test_highway_toll.py completes the three starter tests and adds further properties, for 14 tests in total. Each test runs up to 500 generated inputs, including text built to look nearly valid so that it gets past split() and reaches the int() conversions.

Property	What it checks	Tests
No crash	Random, nearly valid and wrong-type input never raises an exception	7
Rejects bad input	Negative, impossible and over-long values must return None	4
Accepts valid input	Valid input still gives the correct result	3

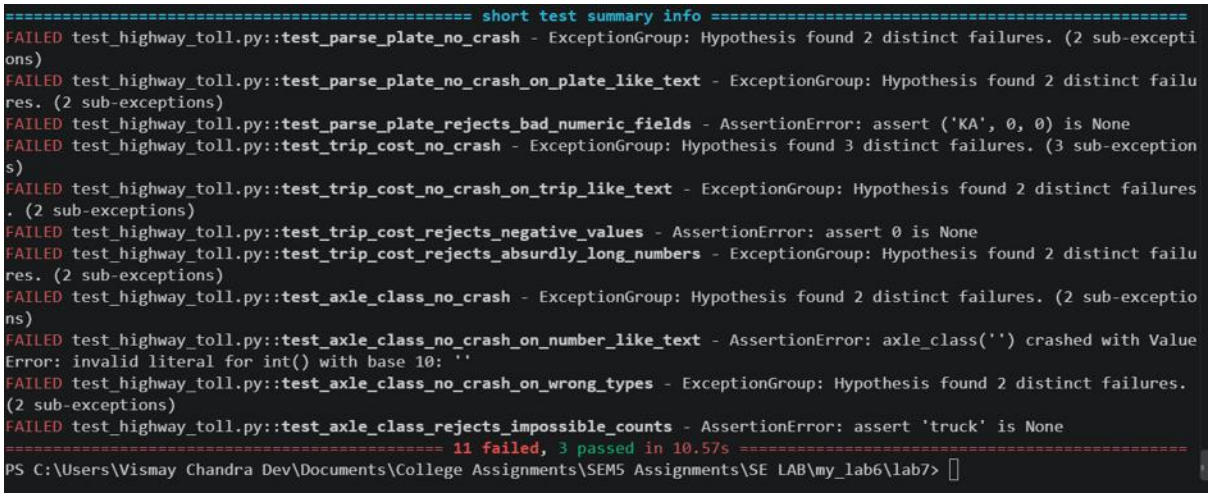


Figure 3: Tests on the buggy code. 11 failed, 3 passed.

4. Bugs Found and Fixes

Invalid input now returns None instead of raising an exception. Each function checks the input type and then matches a strict regular expression before converting anything, because int() alone also accepts values such as "+5" and "1_000". The main program prints a clear message when a function returns None.

Function	Bug in the original code	Fix
parse_plate	Crashes on None, the wrong number of parts and non-numeric fields. Accepts KA-00-AB-0000, returning ('KA', 0, 0).	Plate must match the KA-05-AB-1234 format. District and number must be at least 1.
trip_cost	Crashes on None, 3150, 1x2x3 and very long numbers. -3x150 returns -450.	Text must be two unsigned numbers of up to 9 digits, separated by x or X.
axle_class	Crashes on None, "abc" and infinity. 1, 0 and -7 axles return truck. 3.7 is cut down to 3.	Accepts only whole numbers from 2 to 99, as an int or as text.

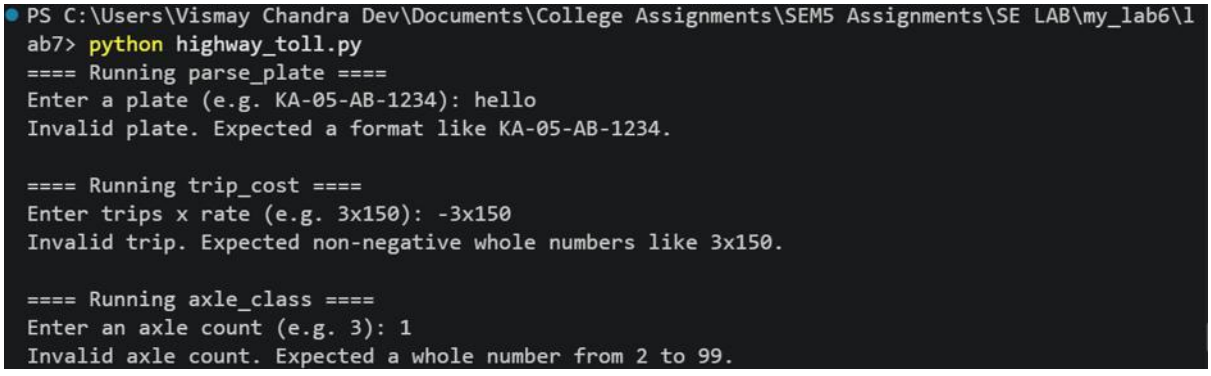


Figure 4: Fixed highway_toll.py showing clear messages for invalid input, with no crash.

5. Re-running the Tests

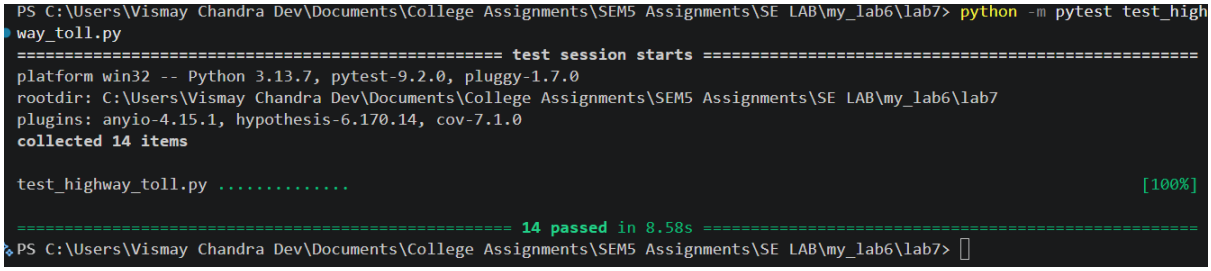


Figure 5: Tests on the fixed code. 14 passed.

6. Results

test_highway_toll.py	Buggy code	Fixed code
Tests failed	11	0
Tests passed	3	14

All the bugs that the fuzz tests found are fixed, and valid input still gives the correct results.